## 2. How do home advantage and opponent strength affect Ashford United’s goals scored and conceded?

We chose this question because Ashford's results were very different in games played home and away. The team won 15 of their 23 home games but only 7 of their 23 away games, which raises a question as to how much of this is down to playing at home, and how much depends on the strength of the opponent. The ashford_matches.csv data contains all 46 league matches, with goals scored, goals conceded, home or away and lastly, the opponent's final league position. We will model goals scored and conceded separately, so we can see if these factors affect how many goals Ashford score, how many they let in, or both.

Goals are counts, so Poisson regression is a suitable model for this statistical analysis. A normal linear model could give negative predictions, which would not make sense for goals. The Poisson model would provide results as rate ratios, which are easy to explain. We can also test whether the home effect changes depending on the opponent's rank by adding an interaction term.

The data also has features that will need checking when we assess the model fit. There is one 8-0 win that could have a large effect on the goals scored model, so we plan to refit the model without it to see how much the results change. Also, goals conceded looks like it has less variation than a Poisson model expects, which we can check with a quasi-Poisson model.